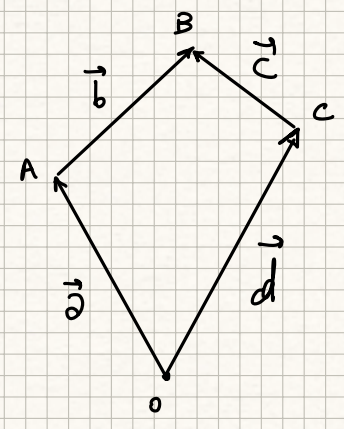
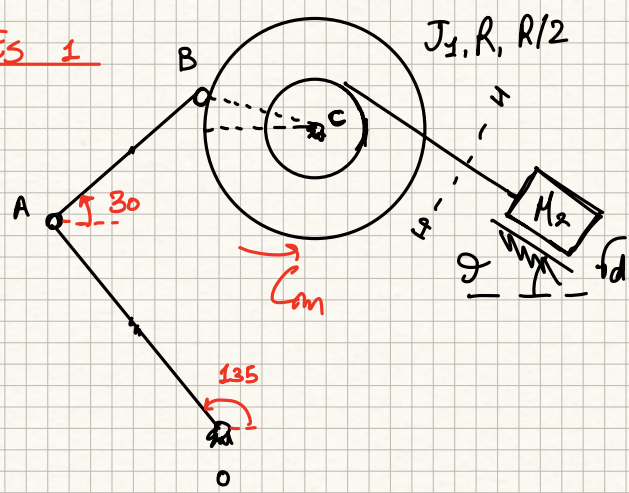


Es 1



Posizione :

$$\begin{cases} a \cos \alpha + b \cos \beta = c \cos \delta + d \cos \delta \\ a \sin \alpha + b \sin \beta = c \sin \delta + d \sin \delta \end{cases}$$

const	a	b	c	d	\delta
var	α	β	δ		

Velocità

$$\begin{cases} -a \dot{\alpha} \sin \alpha - b \dot{\beta} \sin \beta = -c \dot{\delta} \sin \delta \\ a \dot{\alpha} \cos \alpha + b \dot{\beta} \cos \beta = c \dot{\delta} \cos \delta \end{cases}$$

① $\vec{v}_4, \vec{a}_4$

② C_m per atto di moto

③ Tiro H-H

$$\begin{bmatrix} b \sin \beta & -c \sin \delta \\ -b \cos \beta & c \cos \delta \end{bmatrix} \begin{Bmatrix} \dot{\beta} \\ \dot{\delta} \end{Bmatrix} = \begin{Bmatrix} -a \dot{\alpha} \sin \alpha \\ a \dot{\alpha} \cos \alpha \end{Bmatrix} \Rightarrow$$

$$\begin{aligned} \dot{\beta} &= -0.2989 \text{ rad/s} \\ \dot{\delta} &= 0.7436 \text{ rad/s} \end{aligned}$$

①

Accelerazione

$$\begin{cases} -a \ddot{\alpha} \sin \alpha - a \dot{\alpha}^2 \cos \alpha - b \ddot{\beta} \sin \beta - b \dot{\beta}^2 \cos \beta = -c \ddot{\delta} \sin \delta - c \dot{\delta}^2 \cos \delta \\ a \ddot{\alpha} \cos \alpha - a \dot{\alpha}^2 \sin \alpha + b \ddot{\beta} \cos \beta - b \dot{\beta}^2 \sin \beta = c \ddot{\delta} \cos \delta - c \dot{\delta}^2 \sin \delta \end{cases}$$

$$\begin{bmatrix} b \sin \beta & -c \sin \delta \\ -b \cos \beta & c \cos \delta \end{bmatrix} \begin{Bmatrix} \ddot{\beta} \\ \ddot{\delta} \end{Bmatrix} = \begin{Bmatrix} -a \ddot{\alpha} \sin \alpha - a \dot{\alpha}^2 \cos \alpha - b \dot{\beta}^2 \cos \beta + c \dot{\delta}^2 \cos \delta \\ a \ddot{\alpha} \cos \alpha - a \dot{\alpha}^2 \sin \alpha - b \dot{\beta}^2 \sin \beta + c \dot{\delta}^2 \sin \delta \end{Bmatrix}$$

$$\ddot{\beta} = -1.0493 \frac{\text{rad}}{\text{s}^2} \quad \ddot{\delta} = 1.7755 \frac{\text{rad}}{\text{s}^2}$$

$$\vec{v}_A = \dot{\alpha} \vec{k} \wedge \partial (\cos \alpha \vec{i} + \sin \alpha \vec{j}) = \dot{\alpha} \partial (\cos \alpha \vec{j} - \sin \alpha \vec{i})$$

$$\begin{aligned} \vec{v}_1 &= \dot{\alpha} \partial (\cos \alpha \vec{j} - \sin \alpha \vec{i}) + \dot{\beta} \vec{k} \wedge \frac{b}{2} (\cos \beta \vec{i} + \sin \beta \vec{j}) = \\ &= (-\dot{\alpha} \partial \sin \alpha - \dot{\beta} \frac{b}{2} \sin \beta) \vec{i} + (\dot{\alpha} \partial \cos \alpha + \dot{\beta} \frac{b}{2} \cos \beta) \vec{j} = \\ &= (-0.3162 \vec{i} - 0.4183 \vec{j}) \text{ m/s} \end{aligned}$$

$$\begin{aligned} \vec{a}_A &= \ddot{\alpha} \partial (\cos \alpha \vec{j} - \sin \alpha \vec{i}) - \dot{\alpha}^2 \partial (\cos \alpha \vec{i} + \sin \alpha \vec{j}) = \\ &= (-\ddot{\alpha} \partial \sin \alpha - \dot{\alpha}^2 \partial \cos \alpha) \vec{i} + (\ddot{\alpha} \partial \cos \alpha - \dot{\alpha}^2 \partial \sin \alpha) \vec{j} \end{aligned}$$

$$\begin{aligned} \vec{a}_2 &= (-\ddot{\alpha} \partial \sin \alpha - \dot{\alpha}^2 \partial \cos \alpha) \vec{i} + (\ddot{\alpha} \partial \cos \alpha - \dot{\alpha}^2 \partial \sin \alpha) \vec{j} + \\ &+ \ddot{\beta} \vec{k} \wedge \frac{b}{2} (\cos \beta \vec{i} + \sin \beta \vec{j}) - \dot{\beta}^2 \frac{b}{2} (\cos \beta \vec{i} + \sin \beta \vec{j}) = \\ &= (-\ddot{\alpha} \partial \sin \alpha - \dot{\alpha}^2 \partial \cos \alpha - \ddot{\beta} \frac{b}{2} \sin \beta - \dot{\beta}^2 \frac{b}{2} \cos \beta) \vec{i} + \\ &+ (\ddot{\alpha} \partial \cos \alpha - \dot{\alpha}^2 \partial \sin \alpha + \ddot{\beta} \frac{b}{2} \cos \beta - \dot{\beta}^2 \frac{b}{2} \sin \beta) \vec{j} = \\ &= (-0.4185 \vec{i} - 1.222 \vec{j}) \text{ m/s}^2 \end{aligned}$$

$$u_2 = \dot{\gamma} \frac{R}{2} = 0.2788 \text{ m/s} \quad a_2 = \ddot{\gamma} \frac{R}{2} \vec{j} = 0.6658 \text{ m/s}^2$$

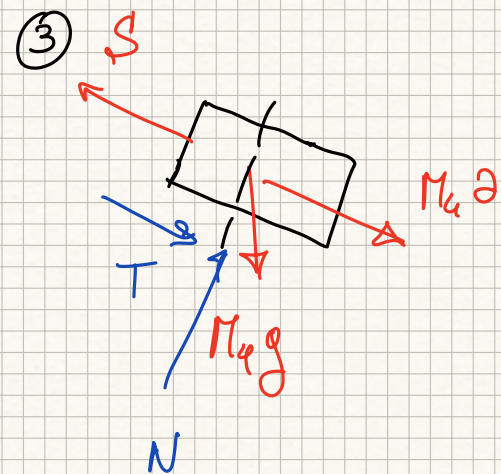
$$\vec{u}_2 = (-0.2415 \vec{i} + 0.1394 \vec{j}) \text{ m/s} \quad \vec{a}_2 = (-0.5766 \vec{i} + 0.3329 \vec{j}) \text{ m/s}^2$$

$$\textcircled{2} \quad \frac{dE_c}{dt} = J_1 \dot{\gamma} \ddot{\gamma} + M_2 u_2 a_2 = 1.588 \text{ W}$$

$$\Sigma W = + C_m \dot{\gamma} - M_2 g \sin \vartheta v_2 - f_d M_2 g \cos \vartheta |v_2| = \frac{dE_c}{dt}$$

$$C_m = \frac{dE_c/dt + M_2 g \sin \vartheta v_2 + f_d M_2 g \cos \vartheta |v_2|}{\dot{\gamma}} = 1452 \text{ Nm}$$

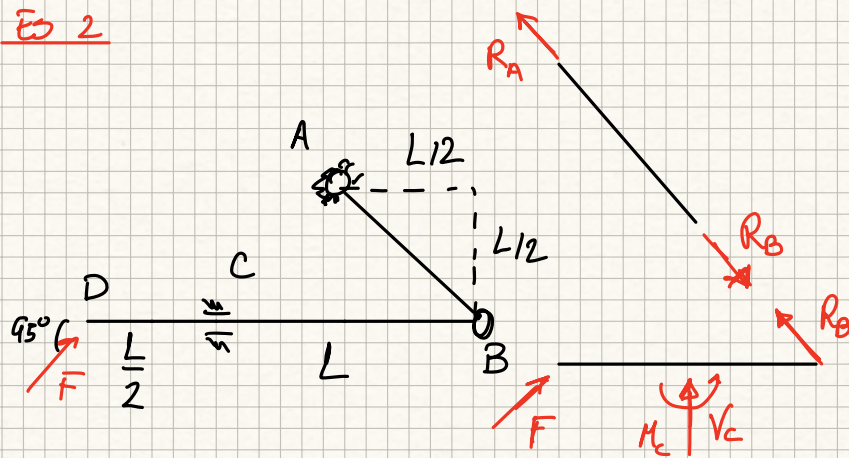
$$\vec{C}_m = (14.52 \vec{u}) \text{ Nm}$$



$$\begin{cases} S = M_3 a + M_3 g \sin \vartheta + T \\ T = f_d N \\ N = M_u g \cos \vartheta \end{cases}$$

$$S = 36.35 \text{ N}$$

ES 2



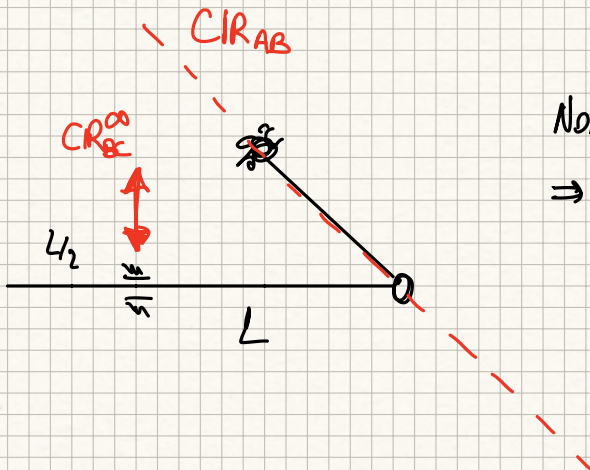
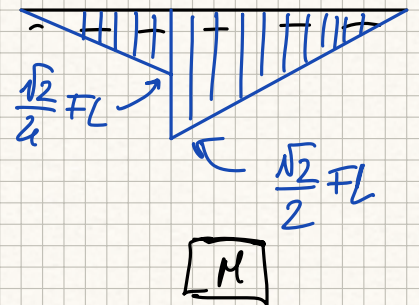
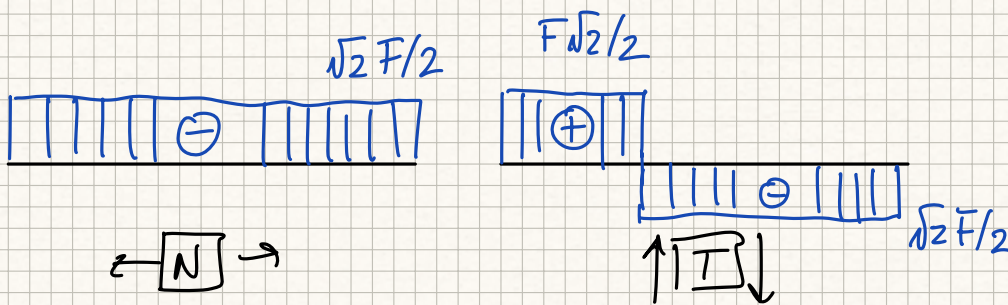
$$R_A = R_B$$

$$-\frac{\sqrt{2}}{2} R_B + F \frac{\sqrt{2}}{2} = 0 \Rightarrow R_B = F$$

$$\frac{\sqrt{2}}{2} R_B + V_C + F \frac{\sqrt{2}}{2} = 0$$

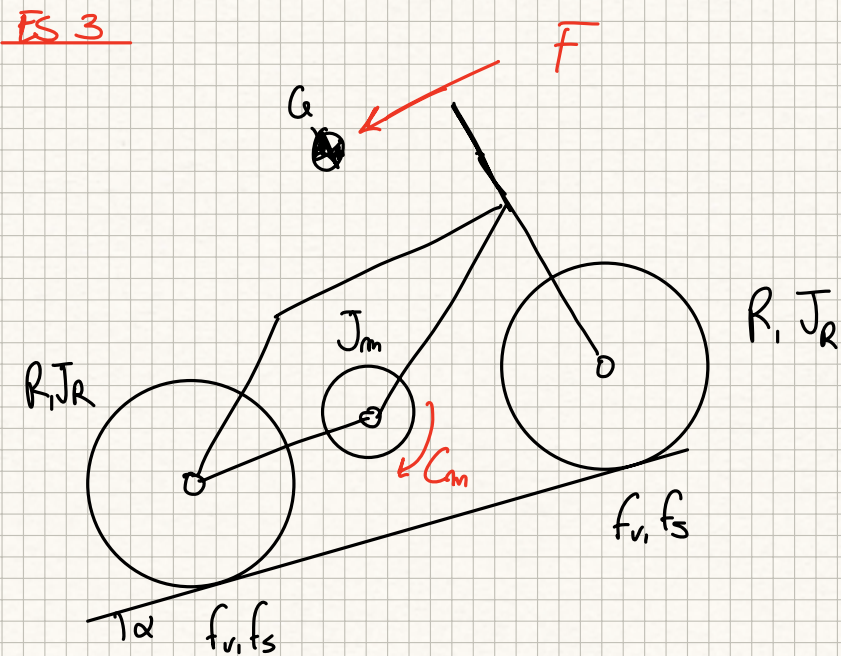
$$V_C = -\sqrt{2} F$$

$$M_C - V_C L - F \frac{\sqrt{2}}{2} \frac{3L}{2} \Rightarrow M_C = -\sqrt{2} FL + FL \frac{3\sqrt{2}}{4} = -\frac{\sqrt{2} FL}{4}$$



Non esiste un unico CR
 $\Rightarrow$ non stabile

ES 3



$$\omega_u = \tau \omega_m = \omega_R$$

$$v = \tau \omega_m R$$

$$W_1 = C_m \omega_m - J_m \dot{\omega}_m \omega_m$$

$$\left. \frac{dE}{dt} \right|_v = M v a + 2 J_R \omega_R \dot{\omega}_R = (M \tau^2 R^2 + 2 J_R \tau^2) \dot{\omega}_m \omega_m$$

$$\Sigma W_v = -M g \sin \alpha v - f_v R M g \cos \alpha \omega_R - F v =$$

$$= [-M g \sin \alpha \tau R - f_v R M g \cos \alpha \tau - F \tau R] \omega_m$$

$$W_2 = C_u^* \omega_m - J_m \dot{\omega}_m \omega_m, \quad J_u^* = 37.6 \text{ kg m}^2, \quad C_u^* = -63.16 \text{ Nm}$$

① C_m @ regime

$$C_u^* < 0 \Rightarrow \text{MOTO DIRETTO}$$

$$M_D C_m + C_u^* = 0 \Rightarrow C_m = 70.18 \text{ Nm}$$

$$\textcircled{2} a = 0.5 \text{ m/s}^2 \Rightarrow \dot{\omega}_m = \frac{a}{\tau R} = 0.833 \text{ rad/s}^2$$

$$W_2 = C_u^* \omega_m - J_m \underbrace{\dot{\omega}_m \omega_m}_{>0} \geq 0?$$

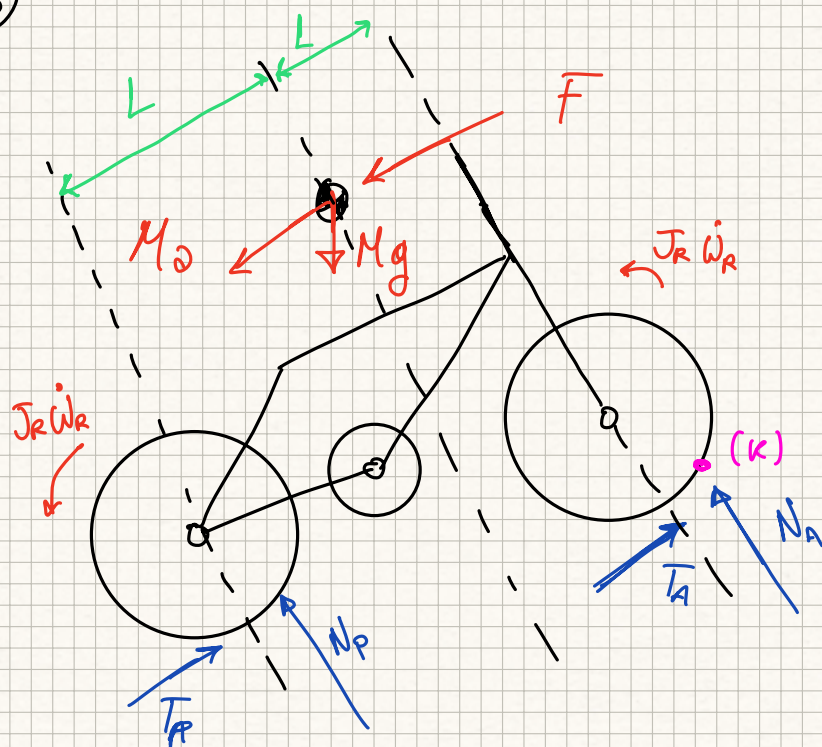
HP: moto diretto

$$\eta_D C_m - \eta_D J_m \dot{\omega}_m + C_u^* - J_u^* \dot{\omega}_m = 0$$

$$\Rightarrow C_m = \frac{\eta_D J_m \dot{\omega}_m + J_u^* \dot{\omega}_m - C_u^*}{\eta_D} = 104.99 \text{ Nm}$$

$$\Rightarrow W_1 = (C_m - J_m \dot{\omega}_m) \omega_m = 104 \omega_m \quad \text{DIRETTO OK}$$

③



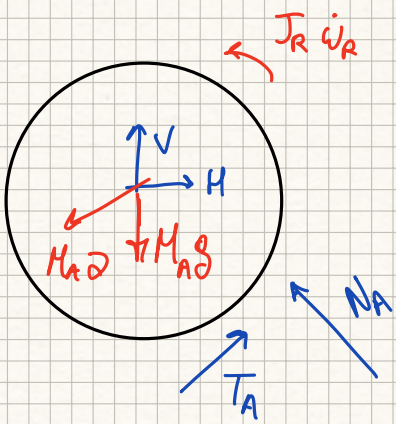
$$\Sigma M_{(K)} = 0$$

$$2J_R \dot{\omega}_R - N_P 2L + M \partial h + F_h + Mg \cos \alpha (L + f_R) + Mg \sin \alpha h = 0$$

$$N_P = \frac{2J_R \partial / R + M \partial h + F_h + Mg \sin \alpha h + Mg \cos \alpha (L + f_R)}{2L}$$

$$N_P = 710.48 \text{ N}$$

$$\Sigma F_L = 0 \Rightarrow Mg \cos \alpha = N_P + N_A \Rightarrow N_A = 266.78 \text{ N}$$



$$J_R \frac{\partial}{R} + T_A R + N_A f_v R = 0$$

$$T_A = -J_R \frac{\partial}{R^2} - N_A f_v = -3.779 \text{ N}$$

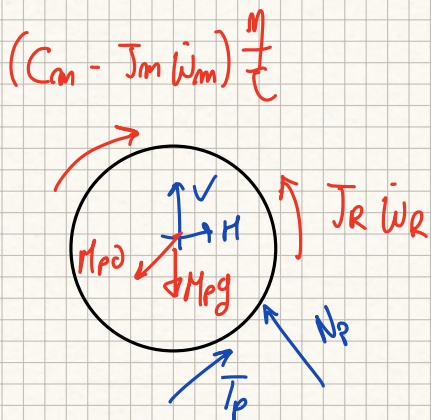
$$T_A \leq f_s N_A = 213.43 \text{ N}$$

$$T_p + T_A - F - M g \sin \alpha - M a = 0$$

$$T_p = F + M g \sin \alpha + M a - T_A \Rightarrow T_p = 149.27 \text{ N}$$

$$T_p \leq f_s N_p = 568.33 \text{ N}$$

Oppure



$$T_p R + N_p f_v R + J_R \dot{\omega}_R - \eta C_m / \tau + \eta \frac{J_m \dot{\omega}_m}{\tau} = 0$$

$$T_p = \eta \frac{C_m}{\tau R} - J_R \frac{\partial}{R^2} - N_p f_v - \eta \frac{J_m \dot{\omega}_m}{\tau R} = 149.27 \text{ N}$$